

Describe your previous research experience or equivalent experience on complex projects, including the level of independence, while working as a member of a research/project team.

My research experience spans computational biophysics and quantum algorithms, where I have been part of multiple projects involving both independent and collaborative work.

As an undergraduate research assistant in the Vermaas lab at the MSU-DOE Plant Research Laboratory, I set up and run molecular dynamics simulations to study biological phenomena at the atomic level. I work independently to prepare simulation systems, run simulations on high performance computing (HPC) clusters, and analyze the resulting data using Python and Tcl scripting, and often integrate feedback from my postdoc mentor and PI to refine my approaches. I am currently working on simulating the effect of butanol and isobutanol on biofuel-producing bacterial membranes containing cyclopropane fatty acid lipids. The aim of doing so is to characterize the disruptive nature of these alcohols on membrane structure and identify potential targets for engineering more robust biofuel-producing strains. My work involves setting up small molecule and lipid force-field parameters, preparing membrane systems, running long-timescale simulations, extracting relevant membrane properties, and visualizing the results for presentations and manuscripts.